

PLANE WAVE PARAMETERS (LOSSLESS)

Angular frequency ω : rad/s; f : Hz

$$\omega = 2\pi f \qquad f = \frac{\omega}{2\pi}$$

Phase velocity m/s ; *nonmagnetic*: $\mu_r = 1$

$$u_p = \frac{c}{\sqrt{\epsilon_r}} \qquad c = 3 \times 10^8 \text{ m/s}$$

Wavenumber k rad/m

$$k = \frac{\omega}{u_p} = \frac{2\pi}{\lambda} = \frac{\omega\sqrt{\epsilon_r}}{c}$$

Wavelength & impedance m & Ω

$$\lambda = \frac{u_p}{f} = \frac{2\pi}{k} \text{ (m)} \qquad \eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \text{ (}\Omega\text{)}, \quad \eta_0 = 377 \text{ }\Omega$$

Order: $\omega \rightarrow u_p \rightarrow k \rightarrow \lambda \rightarrow \eta$. Base: $\mathbf{E} = E_0 \cos(\omega t \pm kx + \phi_0)\hat{e}$.

LOSSLESS PLANE WAVE — E AND H

General E

$$\mathbf{E} = E_0 \cos(\omega t - k \hat{k} \cdot \mathbf{r} + \phi_0) \hat{e}$$

$\hat{k}$ = direction of travel (unit vector), e.g. $+y \Rightarrow \hat{k} = \hat{y}$.
 k = wavenumber (rad/m): how fast phase changes in space; $k = \omega\sqrt{\epsilon_r}/c = 2\pi/\lambda$.
 ϕ_0 = initial phase constant (rad): shift set by given condition at known t , position.
 $\hat{e}$ = polarization direction of $\mathbf{E}$ (unit vector), e.g. E along $x \Rightarrow \hat{e} = \hat{x}$.
 $\mathbf{r} = x\hat{x} + y\hat{y} + z\hat{z}$, so $\hat{k} \cdot \mathbf{r}$ picks the axis, e.g. $\hat{k} = \hat{y} \Rightarrow \hat{k} \cdot \mathbf{r} = y$.
H from **E** $\hat{k}$, $\hat{E}$, $\hat{H}$ form right-hand set

$$\mathbf{H} = \frac{1}{\eta} (\hat{k} \times \mathbf{E})$$

Find ϕ_0 given $E = E_1$ at $t = 0$, position r_0

$$E_0 \cos(-kr_0 + \phi_0) = E_1 \Rightarrow \phi_0 = \cos^{-1}\left(\frac{E_1}{E_0}\right) + kr_0$$

$\cos^{-1}$ gives $[0, \pi]$; take ϕ_0 in that range.

DIRECTION OF PROPAGATION

Arg. contains	$\hat{k}$	Arg. contains	$\hat{k}$
$\omega t - kz$	$+\hat{z}$	$\omega t + kz$	$-\hat{z}$
$\omega t - ky$	$+\hat{y}$	$\omega t + ky$	$-\hat{y}$
$\omega t - kx$	$+\hat{x}$	$\omega t + kx$	$-\hat{x}$

Minus before k -axis $\Rightarrow$ + direction. Plus $\Rightarrow$ - direction.

$\hat{k} \times \hat{E}$ — FIND $\hat{H}$ DIRECTION

$\hat{k}$	$\hat{E}$	$\hat{H}$	$\hat{k}$	$\hat{E}$	$\hat{H}$
$+\hat{z}$	$\hat{x}$	$+\hat{y}$	$+\hat{z}$	$\hat{y}$	$-\hat{x}$
$+\hat{y}$	$\hat{x}$	$-\hat{z}$	$+\hat{y}$	$\hat{z}$	$+\hat{x}$
$+\hat{x}$	$\hat{y}$	$+\hat{z}$	$+\hat{x}$	$\hat{z}$	$-\hat{y}$
$-\hat{x}$	$\hat{y}$	$-\hat{z}$	$-\hat{x}$	$\hat{z}$	$+\hat{y}$
$-\hat{y}$	$\hat{x}$	$+\hat{z}$	$-\hat{z}$	$\hat{x}$	$-\hat{y}$

Cyclic: $\hat{x} \times \hat{y} = \hat{z}$, $\hat{y} \times \hat{z} = \hat{x}$, $\hat{z} \times \hat{x} = \hat{y}$. Reverse $\Rightarrow$ flip sign.

TRIG PHASE SHIFTS ($\theta \equiv \omega t - kz$)

Input	Result	Input	Result
$\sin(\theta + \pi/2)$	$\cos \theta$	$\cos(\theta + \pi/2)$	$-\sin \theta$
$\sin(\theta - \pi/2)$	$-\cos \theta$	$\cos(\theta - \pi/2)$	$\sin \theta$
$\sin(\theta \pm \pi)$	$-\sin \theta$	$\cos(\theta \pm \pi)$	$-\cos \theta$

WAVE PARAMS

$\omega = 2\pi f$ — ang. freq (rad/s)
 $k = \omega\sqrt{\epsilon_r}/c$ — wavenumber (rad/m)
 $\lambda = 2\pi/k$ — wavelength (m)
 $u_p = c/\sqrt{\epsilon_r}$ — phase vel. (m/s)
 $\eta = \eta_0/\sqrt{\epsilon_r}$ — impedance (Ω)
 $c = 3 \times 10^8$ m/s
 $\eta_0 = 377 \Omega$ (free space)

POLARIZATION

a_x, a_y — amplitudes (V/m)
 δ — phase difference (rad)
 $\psi \in [0, \pi)$ — tilt angle
 $\chi \in [-45^\circ, 45^\circ]$ — ellipticity
 $\chi > 0$: RHEP; $\chi < 0$: LHEP
 $|\chi| = 45^\circ$: circular; $\chi = 0$: linear

POLARIZATION — GENERAL WAVE

$$\mathbf{E}(z, t) = a_x \cos(\omega t - kz) \hat{x} + a_y \cos(\omega t - kz + \delta) \hat{y} \quad (\text{V/m})$$

Tilt angle ψ (major axis from $\hat{x}$) $\psi \in [0, \pi)$ rad

$$\tan(2\psi) = \frac{2a_x a_y \cos \delta}{a_x^2 - a_y^2} \quad (\psi \text{ in rad})$$

Ellipticity angle χ $\chi \in [-\pi/4, \pi/4]$ rad

$$\sin(2\chi) = \frac{2a_x a_y \sin \delta}{a_x^2 + a_y^2} \quad (\chi \text{ in rad})$$

Peak amplitude $|\mathbf{E}|_{\text{peak}}$ V/m, linear pol. only

$$|\mathbf{E}|_{\text{peak}} = \sqrt{a_x^2 + a_y^2} \text{ (V/m)}$$

ψ ORIENT. & POL. TYPE

ψ = orientation angle of the ellipse.	
ψ (rad)	Ellipse Orientation
0	Horizontal ($\rightarrow$), aligned with $\hat{x}$
$(0, \pi/4)$	Tilted — horiz side
$\pi/4$	45° diagonal
$(\pi/4, \pi/2)$	Tilted — vert side
$\pi/2$	Vertical ($\uparrow$), aligned with $\hat{y}$

χ SHAPE & HANDEDNESS

χ describes the shape of the ellipse and rotation direction.		
χ (rad)	Polarization Type	Handedness
0	Linear	None
$(0, \pi/4)$	RHEP	RH (CCW)
$\pi/4$	RHC	RH (CCW)
$-\pi/4$	LHC	LH (CW)
$(-\pi/4, 0)$	LHEP	LH (CW)

QUADRANT RULE FOR $\tan(2\psi)$

$N = 2a_x a_y \cos \delta, \quad D = a_x^2 - a_y^2$			
N	D	2ψ (deg)	2ψ (rad)
+	+	$\arctan(N/D)$	$\arctan(N/D)$
+	-	$180^\circ - \arctan N/D $	$\pi - \arctan N/D $
-	-	$180^\circ + \arctan N/D $	$\pi + \arctan N/D $
-	+	$360^\circ - \arctan N/D $	$2\pi - \arctan N/D $

$\psi = 2\psi/2$. $D = 0 \Rightarrow 2\psi = \frac{\pi}{2} \Rightarrow \psi = \frac{\pi}{4}$.
Convert: $\theta_{\text{rad}} = \theta_\circ \times \frac{\pi}{180}$; $\theta_\circ = \theta_{\text{rad}} \times \frac{180}{\pi}$.

PHASOR $\leftrightarrow$ TIME DOMAIN

Phasor coeff	Time domain (add $e^{-\alpha z}$)
A (real)	$A \cos(\omega t - \beta z)$
$-jA$	$A \sin(\omega t - \beta z)$
$+jA$	$-A \sin(\omega t - \beta z)$
$Ae^{j\phi}$	$A \cos(\omega t - \beta z + \phi)$
$A + jB$	$ Z \cos(\omega t - \beta z + \theta)$

$|Z| = \sqrt{A^2 + B^2}$, $\theta = \arctan(B/A)$ (check quadrant)

LOSSY MEDIA — CLASSIFICATION

Loss tangent $\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$		
ϵ''/ϵ'	Type	Use
> 100	Good conductor	GC approx
0.01 to 100	Quasi-conductor	Exact eqs
< 0.01	Low-loss dielectric	LD approx
<i>Non-magnetic ($\mu_r = 1$): $\mu = \mu_0$. $\epsilon' = \epsilon_r \epsilon_0$ ($\epsilon_r = 1$: vacuum/air).</i>		

EXACT FORMULAS (QUASI-CONDUCTOR)

Let $X = \sqrt{1 + (\sigma/\omega\epsilon)^2} = \sqrt{1 + (\epsilon''/\epsilon')^2}$, $\epsilon'' = \sigma/\omega$, $\epsilon = \epsilon_r \epsilon_0$, $\mu = \mu_0$
 α and β

$$\alpha = \omega \sqrt{\frac{\mu\epsilon}{2}} \sqrt{X - 1} \text{ (Np/m)}, \quad \beta = \omega \sqrt{\frac{\mu\epsilon}{2}} \sqrt{X + 1} \text{ (rad/m)}$$
$$\lambda = \frac{2\pi}{\beta} \text{ (m)}, \quad u_p = \frac{\omega}{\beta} \text{ (m/s)}$$
$$\eta_c = \frac{\eta}{X^{1/2}} \angle \theta_\eta \text{ (}\Omega\text{)}, \quad \theta_\eta = \frac{1}{2} \arctan \frac{\sigma}{\omega\epsilon} \text{ (deg)}$$

where $\eta = \eta_0/\sqrt{\epsilon_r}$.

GOOD CONDUCTOR ($\sigma/\omega\epsilon \gg 100$)

$$\alpha \approx \omega \sqrt{\frac{\mu\epsilon''}{2}} = \omega \sqrt{\frac{\mu\sigma}{2\omega}} = \sqrt{\pi f \mu \sigma} \text{ (Np/m)}$$
$$\beta \approx \alpha \approx \sqrt{\pi f \mu \sigma} \text{ (rad/m)}$$
$$u_p = \omega/\beta \text{ (m/s)}, \quad \lambda = 2\pi/\beta \text{ (m)}$$
$$\eta_c \approx \sqrt{j \frac{\mu}{\epsilon''}} = (1 + j) \sqrt{\frac{\pi f \mu}{\sigma}} \approx (1 + j) \frac{\alpha}{\omega} \text{ (}\Omega\text{)}$$

LOW-LOSS DIELECTRIC ($\sigma/\omega\epsilon \ll 0.01$)

$$\alpha \approx \omega \frac{\epsilon''}{2} \sqrt{\frac{\mu}{\epsilon'}} = \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon'}} = \frac{\sigma \eta}{2} \text{ (Np/m)}$$
$$\beta \approx \omega \sqrt{\mu\epsilon'} = \frac{\omega}{c} \sqrt{\epsilon_r} \text{ (rad/m)}, \quad u_p \approx \frac{c}{\sqrt{\epsilon_r}} \text{ (m/s)}$$
$$\eta_c \approx \sqrt{\frac{\mu}{\epsilon'}} = \eta \text{ (}\Omega\text{), } \angle \approx 0^\circ$$

$\hat{\mathbf{H}}$ PHASOR $\rightarrow \hat{\mathbf{E}}$ IN LOSSY MEDIA

Step 1 — read α, β

$$\hat{\mathbf{H}} \propto e^{-\alpha y} e^{-j\beta y} \Rightarrow \gamma = \alpha + j\beta$$

Step 2 — η_c (nonmagnetic: $\mu_r = 1$)

$$\eta_c = \frac{j\omega\mu_0}{\gamma} = \frac{\omega\mu_0(\beta + j\alpha)}{\alpha^2 + \beta^2}$$

Step 3 — find $\tilde{\mathbf{E}}$

$$\tilde{\mathbf{E}} = -\eta_c (\hat{k} \times \hat{\mathbf{H}})$$

Step 4 — time domain

$$Ae^{j\phi} e^{-\alpha y} \xrightarrow{\text{Re}\{ \cdot e^{j\omega t} \}} Ae^{-\alpha y} \cos(\omega t - \beta y + \phi)$$

Convert each component $A + jB \rightarrow |Z| \angle \theta$ first.

AVERAGE POWER DENSITY (POYNTING)

General W/m^2

$$\mathbf{S}_{\text{av}} = \frac{1}{2} \text{Re}\{\tilde{\mathbf{E}} \times \hat{\mathbf{H}}^*\}$$

Lossless shortcut $|E_0|^2 = a_x^2 + a_y^2 + a_z^2$

$$\mathbf{S}_{\text{av}} = \frac{|E_0|^2}{2\eta} \hat{k}$$

Direction = $\hat{k}$. If wave goes in $-\hat{x}$, then $\mathbf{S}_{\text{av}}$ is a negative x -component.

CIRCULAR WAVES

$\tilde{\mathbf{E}}_R = a_R(\hat{x} + e^{-j\pi/2}\hat{y})e^{-jkz} = a_R(\hat{x} - j\hat{y})e^{-jkz}$
 $\tilde{\mathbf{E}}_L = a_L(\hat{x} + e^{+j\pi/2}\hat{y})e^{-jkz} = a_L(\hat{x} + j\hat{y})e^{-jkz}$
 a_R, a_L, a_x in V/m; k in rad/m
Linear: $a_R = a_L = a_x/2$
 $-j = e^{-j\pi/2}$: $\cos \rightarrow \sin$
 $+j = e^{+j\pi/2}$: $\cos \rightarrow -\sin$

LOSSY MEDIA

α — atten. const (Np/m)
 β — phase const (rad/m)
 $\gamma = \alpha + j\beta$ — prop. const (m^{-1})
 η_c — complex impedance (Ω)
 $\epsilon' = \epsilon_r \epsilon_0$; $\epsilon'' = \sigma/\omega$ (F/m)
 μ_r — rel. permeability ($\mu_r = 1$: nonmagnetic)
 $\mu = \mu_r \mu_0$; $\mu_0 = 4\pi \times 10^{-7}$ H/m
 $\delta_s = 1/\alpha$ — skin depth (m)

POLARIZATION

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USEFUL NUMBERS

$\sqrt{2.56} = 1.6$; $\sqrt{9} = 3$
 $\sqrt{2.5} \approx 1.581$; $\sqrt{2} \approx 1.414$
 $\arccos(2/3) \approx 48.2^\circ = 0.84$ rad
 $\arctan(4/3) \approx 53.1^\circ = 0.927$ rad
 $\arctan(3/4) \approx 36.9^\circ = 0.644$ rad
 $\arctan(24/7) \approx 73.7^\circ = 1.287$ rad